Experiment Number: 3 MATLAB Analysis of Spectrum Bands, Propagation Loss and Cellular Interference

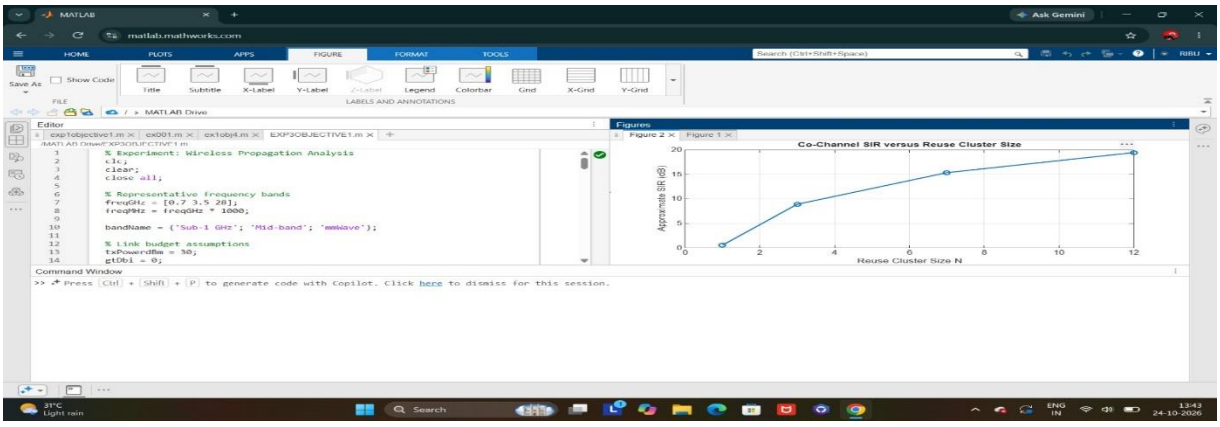
PROGRAM:

```
% Experiment 3: Illustrative Propagation and Frequency Reuse

clear;
clc; close
all;
%% 1. Representative frequency bands freqGHz =
[0.7; 3.5; 28]; freqMHz = freqGHz * 1000; bandName
= ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30; gtDbi = 0; grDbi = 0; distanceKm
= logspace(log10(0.1), log10(2), 200);
distanceAtPointKm = 0.5;
%% 2. Free-space path loss model
% d in km, f in MHz fspl
= @(fMHz, dKm) ...
   32.44 + 20*log10(fMHz) + 20*log10(dKm); lossAtPoint
= fspl(freqMHz, distanceAtPointKm); rxAtPoint =
txPowerdBm + gtDbi + grDbi - lossAtPoint; linkTable =
table( ...
   bandName, freqGHz, lossAtPoint, rxAtPoint, ...
   'VariableNames', ...
   {'Band', 'Frequency_GHz', 'FSPL_dB', 'EstimatedRx_dBm'});
disp('LINK BUDGET AT 0.5 km'); disp(linkTable);
%% 3. Plot free-space path loss and received power
figure('Color', 'w', 'Name', 'Propagation Analysis'); tiledlayout(1,
2, ...
   'TileSpacing', 'compact', ...
   'Padding', 'compact'); % Free-space path loss
nexttile; hold on; grid on; for k = 1:numel(freqMHz)
plot(distanceKm, fspl(freqMHz(k), distanceKm), ...
      'LineWidth', 1.5);
end
set(gca, 'XScale', 'log'); xlabel('Distance
(km)'); ylabel('Path loss (dB)');
legend(bandName, 'Location', 'northwest');
title('Free-Space Path Loss'); hold off;
% Estimated received power nexttile; hold
on; grid on; for k = 1:numel(freqMHz)
rx = txPowerdBm + gtDbi + grDbi ...
- fspl(freqMHz(k), distanceKm);
```

```
plot(distanceKm, rx, 'LineWidth', 1.5); end
set(gca, 'XScale', 'log'); xlabel('Distance
(km)'); ylabel('Estimated received power
(dBm)'); legend(bandName, 'Location',
'southwest'); title('Illustrative Received
Power'); hold off;
%% 4. Simplified co-channel frequency reuse
cellRadiusKm = 0.5; reuseN = [1; 3; 7; 12];
pathLossExponent = 3.5; numberOfInterferers = 6;
% Hexagonal cellular geometry reuseDistance =
sqrt(3 .* reuseN) .* cellRadiusKm; % Approximate
SIR:
% SIR = (D/R)^n / number of equal-distance interferers
sirLinear = (reuseDistance ./ cellRadiusKm) ...     .^
pathLossExponent / numberOfInterferers; sirDb =
10*log10(sirLinear); reuseTable = table( ...
   reuseN, reuseDistance, sirDb, ...
   'VariableNames', ...
   {'ReuseClusterN', 'ReuseDistance_km', 'ApproxSIR_dB'});
disp('FREQUENCY REUSE ANALYSIS');
disp(reuseTable);
%% 5. Plot SIR versus reuse cluster size
figure('Color', 'w'); plot(reuseN, sirDb,
'o-', 'LineWidth', 1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR
(dB)'); title('Co-Channel SIR versus Reuse Cluster Size');
%% 6. Display received power summary fprintf('\nAt
%.2f km, estimated received powers are:\n', ...
   distanceAtPointKm);
rxTable     =     table(     ...
bandName, rxAtPoint, ...
   'VariableNames', {'Band', 'Rx_dBm'}); disp(rxTable);
```

OUTPUT:

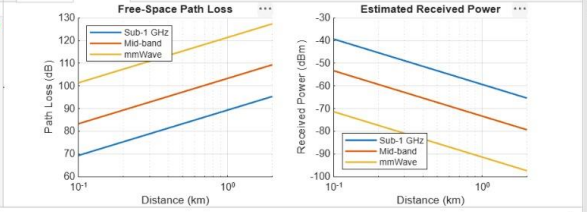


```
% Experiment: Wireless Propagation Analysis
clc;
clear;
close all;

% Representative frequency bands
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;

bandName = {'Sub-1 GHz'; 'Mid-band'; 'mmWave'};

% Link budget assumptions
txPowerdBm = 30;
gtDbi = 0;
```




```
>> Press Ctrl + Shift + P to generate code with Copilot. Click here to dismiss for this session.
```

```
      Band          Frequency_GHz    FSPL_dB    EstimatedRx_dBm
    ____________    _____________    _______    _______________

    "Sub-1 GHz"           0.7          83.321        -53.321
    "Mid-band"            3.5          97.301        -67.301
    "mmWave"               28          115.36        -85.363

    ReuseClusterN    ApproxInterfererDistance_km    ApproxSIR_dB
    _____________    ___________________________    ____________

            1                   0.86603                0.56811
            3                       1.5                8.9177
            7                    2.2913                15.357
           12                         3                19.454

At 0.50 km, estimated received powers are:
      Band          Rx_dBm
    ____________    _______

    "Sub-1 GHz"     -53.321
    "Mid-band"      -67.301
    "mmWave"        -85.363

>>
```